HERMES

— WHERE WASTE TAKES FLIGHT —

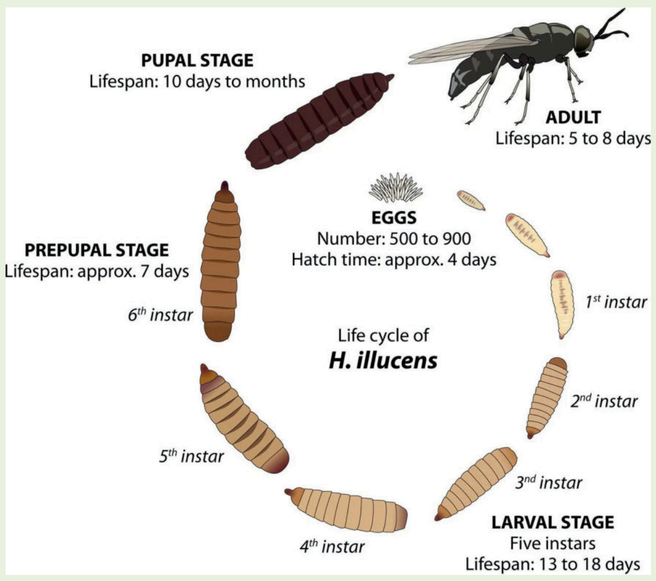
BY GROUP 21

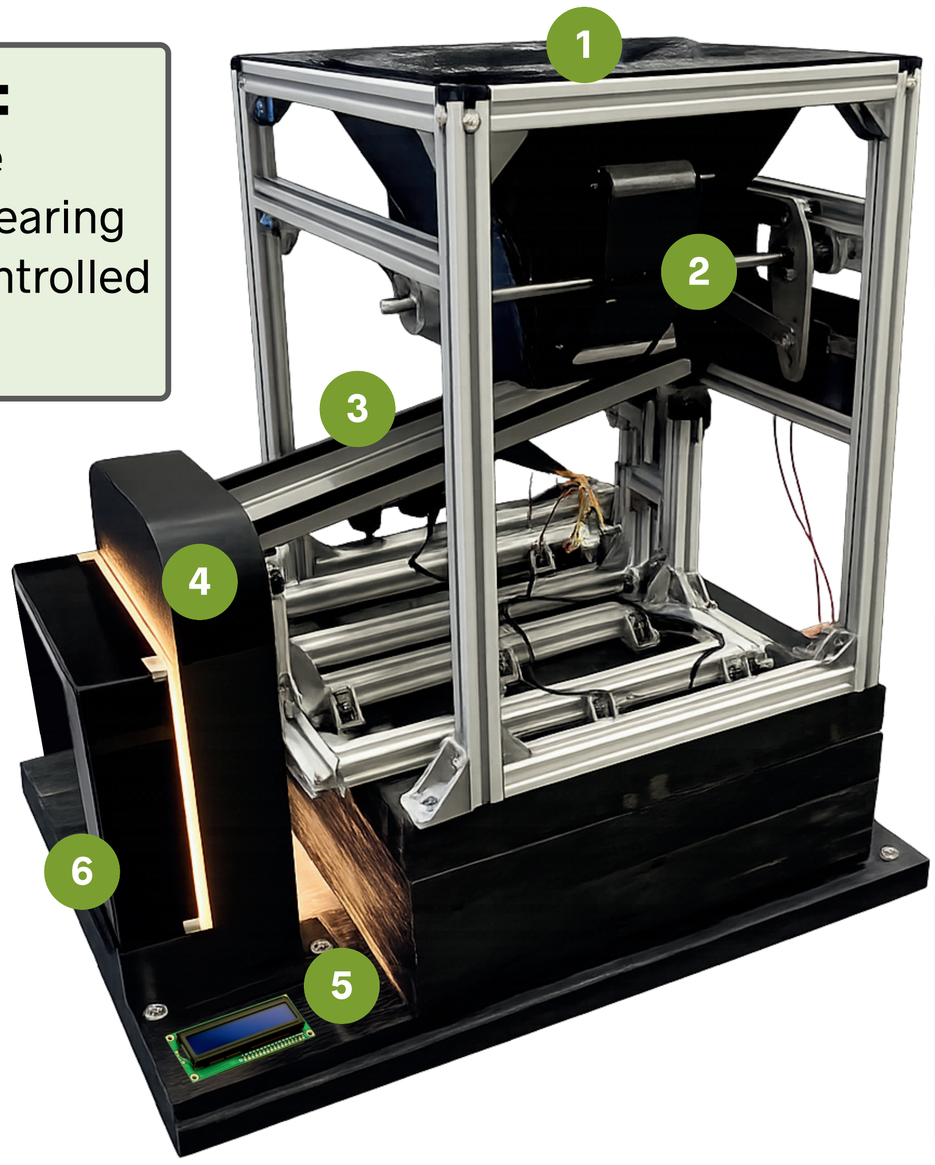
Joshua Loo Yong Jie
Phelia Hoe Ying Qian
Puah Yi Da
Bryan Wilson Chandra
Too Zhan Yue
Benedict Julian Danandjaja Antoni
Sng Ming Xiang
Kng Jia Qing

OUR PROBLEM STATEMENT:

How might we **automate the accurate measurement of BSF neonates** into rearing trays so that each tray starts with a controlled neonate-to-substrate ratio?

LIFE CYCLE OF A BLACK SOLDIER FLY (BSF)





1 Egg Hatching Frame



Neonates hatch across four days and drop continuously, instead of one manual collection a day.

2 Deforming Funnel



A 3D-printed TPU frustum, tapped by a slider-crank paddle so neonates never stick to the wall.

3 Vibrating Ramp



Spring-mounted DC vibration motors declump the stream and meter it into the drop zone.

4 Event-Based Camera



An OpenMV GENX320 detects motion against an LED backlight, counting and posting a live total.

5 LED Screen



Displays live count of neonates, counted by event based camera

6 Rearing Tray



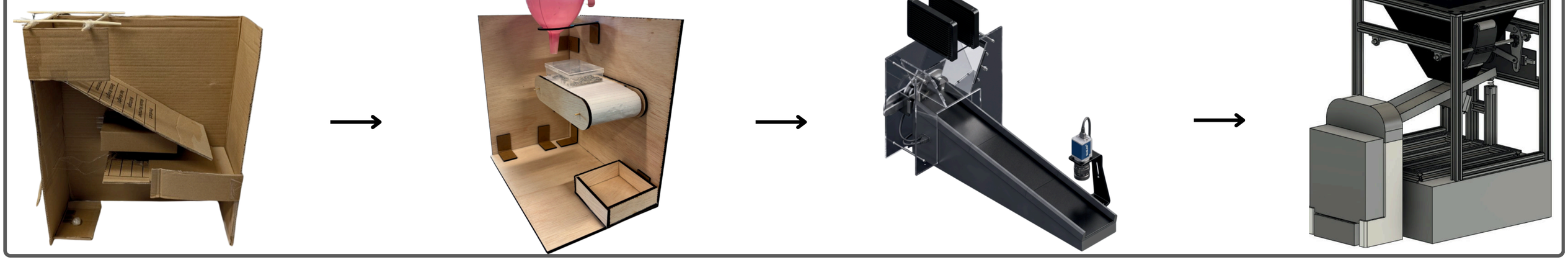
A known count lands in the tray, giving the controlled substrate-to-neonate ratio.

98.6%
Counting Accuracy for Drops of 10
SD: σ = 2.20, against 1000 hand-counted drops. Requirement was 90%

100%
Survival and Flow-through
Every live neonate cleared all three subsystems alive.

< 0.4%
Of Commercial Cost
$1,120.20 est per-unit build, entirely off-the-shelf and 3D-printed.

OUR PROTOTYPING JOURNEY: EVOLUTION OF HERMES

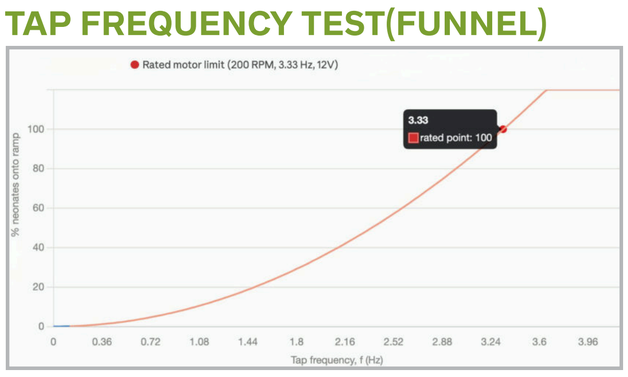


TESTS CONDUCTED

FUNNEL	Effectiveness of Different Funnel Materials
	Funnel Shape Experimentation
	Survivability Test for Live neonates
RAMP	Optimal angle test
	Vibration Frequency Test (optimal voltage test)
	Survivability Test for Live neonates
CAMERA	Algorithm Optimisation
	Ground Truth Testing with filament drops
	Counting of live neonates

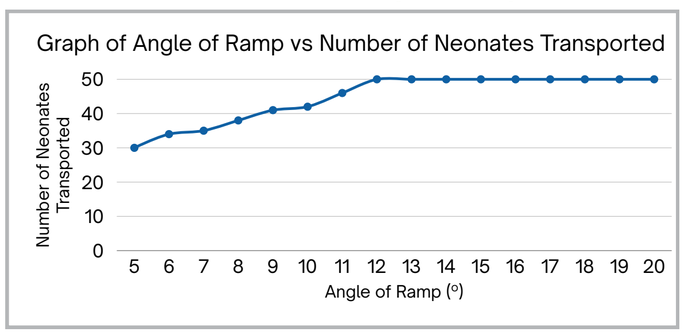
RESULTS & GRAPHS

TAP FREQUENCY TEST(FUNNEL)



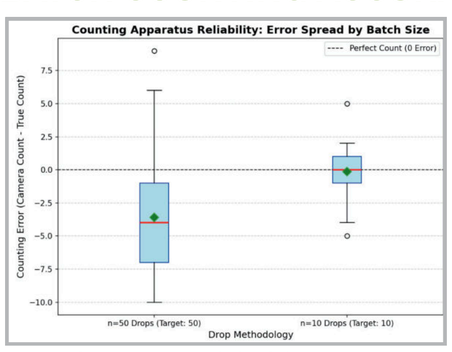
Declumping success increases with tap frequency, reaching 100% at 3.33Hz.

ANGLE OPTIMISATION (RAMP)



Minimum ramp angle = 12°

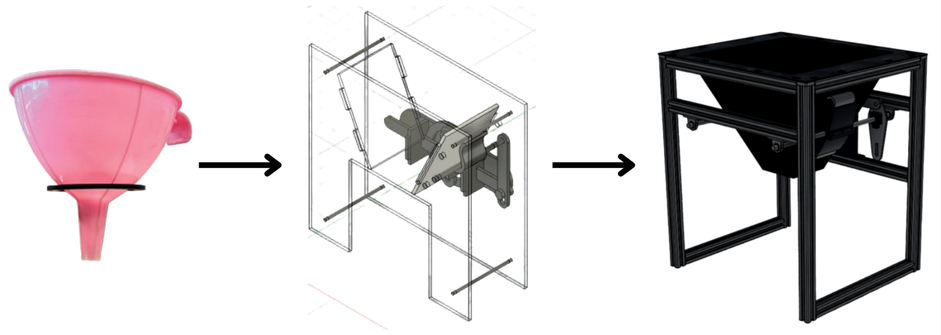
BATCH COUNTING ACCURACY (CAMERA)



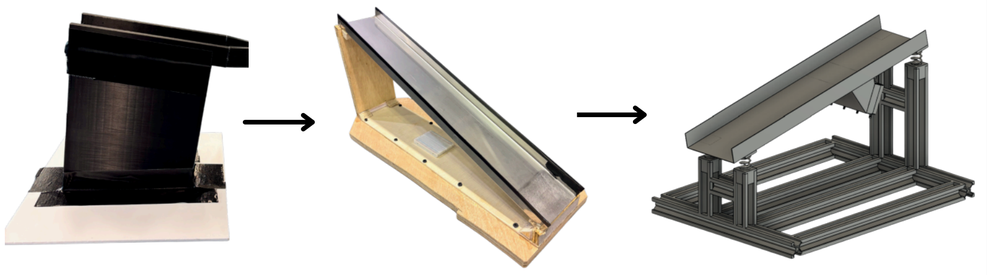
n=50 Drops: 92.9% Avg. Accuracy | SD = 5.56
n=10 Drops: 98.6% Avg. Accuracy | SD = 2.20

ITERATION HISTORY

FUNNEL



RAMP



CAMERA

